

```
#load datasets
iris = load_iris()
X = iris.data
y = iris.target
```

```
[7.9, 3.8, 6.4, 2. ],  
[6.4, 2.8, 5.6, 2.2],  
[6.3, 2.8, 5.1, 1.5],  
[6.1, 2.6, 5.6, 1.4],  
[7.7, 3. , 6.1, 2.3],  
[6.3, 3.4, 5.6, 2.4],  
[6.4, 3.1, 5.5, 1.8],  
[6. , 3. , 4.8, 1.8],  
[6.9, 3.1, 5.4, 2.1],  
[6.7, 3.1, 5.6, 2.4],  
[6.9, 3.1, 5.1, 2.3],  
[5.8, 2.7, 5.1, 1.9],  
[6.8, 3.2, 5.9, 2.3],  
[6.7, 3.3, 5.7, 2.5],  
[6.7, 3. , 5.2, 2.3],  
[6.3, 2.5, 5. , 1.9],  
[6.5, 3. , 5.2, 2. ],  
[6.2, 3.4, 5.4, 2.3],  
[5.9, 3. , 5.1, 1.8]]),  
  
'target': array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]),  
  
'frame': None,  
  
'target_names': array(['setosa', 'versicolor', 'virginica'], dtype='<U10'),  
  
'DESCR': '.. _iris_dataset:\n\nIris plants dataset\n-----\n\n**Data Set Characteristics:**\n\nNumber of  
Instances: 50 (50 in each of three classes)\nNumber of Attributes: 4 numeric, predictive attributes and the  
class\nAttribute Information:\n    - sepal length in cm\n    - sepal width in cm\n    - petal length in cm\n    - petal  
width in cm\n    - class:\n        - Iris-Setosa\n        - Iris-Versicolour\n        - Iris-\nVirginica\nSummary Statistics:\n=====\nMin  
Max Mean SD Class Correlation\nsepal length: 4.3  
7.9 5.84 0.83 0.7826\nsepal width: 2.0 4.4 3.05 0.43 -0.4194\npetal length: 1.0 6.9 3.76 1.76  
0.9490 (high!)\npetal width: 0.1 2.5 1.20 0.76 0.9565 (high!) \n===== \n\nMissing Attribute Values: None\nClass Distribution: 33.3% for each of 3 classes.\nCreator:  
R.A. Fisher\nDonor: Michael Marshall (MARSHALL@PLU.io.arc.nasa.gov)\nDate: July, 1988\nThe famous Iris database,\nfirst used by Sir R.A. Fisher. The dataset is taken from Fisher's paper. Note that it's the same as in R, but not as  
in the UCI Machine Learning Repository, which has two wrong data points. This is perhaps the best known database to  
be found in the pattern recognition literature. Fisher's paper is a classic in the field and is referenced frequently  
to this day. (See Duda & Hart, for example.) The data set contains 3 classes of 50 instances each, where each class  
refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly  
separable from each other.\n.. dropdown:: References\n- Fisher, R.A. "The use of multiple measurements in taxonomic problems"\nAnnual Eugenics, 7, Part II, 179-188 (1936); also in "Contributions to Mathematical Statistics" (John Wiley, NY, 1950).\n- Duda, R.O., & Hart, P.E. (1973) Pattern Classification and Scene Analysis. (Q327.D83) John Wiley & Sons. ISBN 0-471-22361-1. See page 218.\n- Dasarthy, B.V. (1980) "Nosing Around the Neighborhood: A New System Structure and Classification Rule for Recognition in Partially Exposed Environments". IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. PAMI-2, No. 1, 67-71.\n- Gates, G.W. (1972) "The Reduced Nearest Neighbor Rule". IEEE Transactions on Information Theory, May 1972, 431-433.\n- See also: 1988 MLC Proceedings, 54-64. Cheeseman et al's AUTOCLASS II conceptual clustering system finds 3 classes in the data.\nMany, many more ...'\n'  
  
'feature_names': ['sepal length (cm)',  
'sepal width (cm)',  
'petal length (cm)',  
'petal width (cm)' ],  
  
'filename': 'iris.csv',  
  
'data module': 'sklearn.datasets.data'
```

```
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
```

Start coding or generate with AI.

```
#predict and evaluate
y_pred = rf.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.2f}")
```

Accuracy: 1.00

```
#predict on new data
new_data = [[5.1, 3.5, 1.4, 0.2]]
new_predictions = rf.predict(new_data)
predicted_class = iris.target_names[new_predictions[0]]
print(f"Predicted species: {predicted_class}")
```

Predicted species: setosa